In [1]:

```
from tensorflow.keras.datasets import fashion_mnist
```

In [2]:

```
(train_x, train_y), (test_x, test_y) = fashion_mnist.load_data()
```

Out:

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-label
s-idx1-ubyte.gz
29515/29515 [==============================] - 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-image
s-idx3-ubyte.gz
26421880/26421880 [==============================] - 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-labels
-idx1-ubyte.gz
5148/5148 [==============================] - 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-images
-idx3-ubyte.gz
4422102/4422102 [==============================] - 0s 0us/step
```

In [3]:

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten, MaxPooling2D, Conv2D
```

In [4]:

```
model = Sequential()
```

In [5]:

```
model.add(Conv2D(filters=64,kernel_size=(3,3),activation='relu',input_shape=(28, 28,
1)))

# Adding maxpooling layer to get max value within a matrix
model.add(MaxPooling2D(pool_size=(2,2)))

model.add(Flatten())
model.add(Dense(128, activation = "relu"))
model.add(Dense(10, activation = "softmax"))
```

In [6]:

```
model.summary()
```

Out:

```
Model: "sequential_2"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 flatten_1 (Flatten)         (None, 784)               0

 dense_8 (Dense)             (None, 128)               100480

 dense_9 (Dense)             (None, 10)                1290

=================================================================
Total params: 101,770
Trainable params: 101,770
Non-trainable params: 0
_________________________________________________________________
```

In [7]:

```
model.compile(optimizer = 'adam', loss = 'sparse_categorical_crossentropy', metrics = ['accuracy'])
```

In [8]:

```
model.fit(train_x.astype(np.float32), train_y.astype(np.float32), epochs = 5, validation_split = 0.2)
```

Out:

```
Epoch 1/5
1500/1500 [==============================] - 7s 5ms/step - loss: 0.4881 - accuracy: 0.8302 - val_loss: 0.5287 - val_accuracy: 0.8273
Epoch 2/5
1500/1500 [==============================] - 6s 4ms/step - loss: 0.4893 - accuracy: 0.8298 - val_loss: 0.5376 - val_accuracy: 0.8243
Epoch 3/5
1500/1500 [==============================] - 7s 5ms/step - loss: 0.4774 - accuracy: 0.8342 - val_loss: 0.5451 - val_accuracy: 0.8282
Epoch 4/5
1500/1500 [==============================] - 6s 4ms/step - loss: 0.4751 - accuracy: 0.8361 - val_loss: 0.5717 - val_accuracy: 0.8299
Epoch 5/5
1500/1500 [==============================] - 7s 5ms/step - loss: 0.4753 - accuracy: 0.8363 - val_loss: 0.5278 - val_accuracy: 0.8255
```

```
<keras.callbacks.History at 0x7fbcee0aceb0>
```

In [9]:

```
loss, acc = model.evaluate(test_x, test_y)
```

Out:

```
313/313 [==============================] - 1s 2ms/step - loss: 0.5724 - accuracy: 0.8171
```

In [10]:

```
labels = ['t_shirt', 'trouser', 'pullover', 'dress', 'coat', 'sandal', 'shirt', 'sne
aker',   'bag', 'ankle_boots']
```

In [11]:

```
predictions = model.predict(test_x[:1])
```

Out:

```
1/1 [==============================] - 0s 73ms/step
```

In [12]:

```
import numpy as np
```

In [13]:

```
label = labels[np.argmax(predictions)]
```

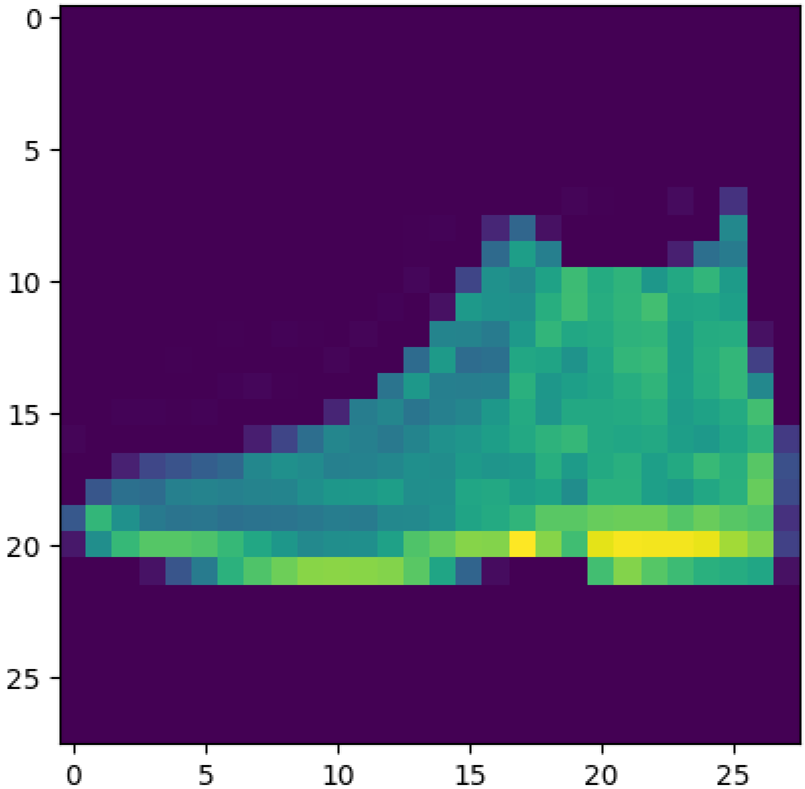
In [14]:

```
import matplotlib.pyplot as plt
print(label)
plt.imshow(test_x[:1][0])
plt.show
```

Out:

```
ankle_boots
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```


0
5
10
15
20
25
0
5
10
15
20
25